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AUTOSAR HIGH-LEVEL DESIGN SPECIFICATION (HLD)

Document ID: HLD-BMS-2026-V2.0

System: Electric Vehicle Battery Management System (BMS) - Fast Charging Upgrade

Version: 2.0 (Updated Architecture)

Classification: ASIL-D

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1. System Architectural Overview

Version 2.0 introduces ultra-fast DC charging support (800V architecture), active cell balancing control, and cybersecurity authentication.

Updated Requirements:

- REQ_BMS_001: The system shall monitor all individual battery cell voltages with an update rate of 50ms.
- REQ_BMS_002: The system shall monitor cell temperatures and trigger cooling when temp exceeds 45°C.
- REQ_BMS_003: The system shall calculate Battery State of Charge (SOC) and State of Health (SOH).
- REQ_BMS_004: The contactor control manager shall open HV contactors upon critical over-current or isolation fault.
- REQ_BMS_005: Thermal management service shall regulate coolant pump flow based on temperature gradients.
- REQ_BMS_006: The system shall execute active cell balancing algorithms during DC fast charging.
- REQ_BMS_007: IsolationMonitor shall continuously evaluate high-voltage chassis insulation resistance.

2. Software Components (SWC) Definition

- CellVoltageMonitor (SensorActuatorSWC):
Acquires cell voltages via ADC / SPI abstraction and publishes cell voltage arrays.
- CellTemperatureMonitor (SensorActuatorSWC):
Reads NTC thermistor sensor arrays across battery modules.
- SOC_SOE_Estimator (ApplicationSWC):
Enhanced multi-model observer estimating SOC, SOH, and maximum fast-charge pulse power.
- ContactorControlManager (ApplicationSWC):
Handles 800V DC fast charge contactor sequencing, AC charging interlocks, and emergency disconnects.
- ThermalManagementService (ServiceSWC):
Advanced chiller regulation supporting rapid pre-conditioning for high-power DC fast charging.
- ActiveCellBalancingManager (ApplicationSWC): [NEW in V2.0]
Controls bi-directional flyback active balancing circuits to equalize cell voltages dynamically.
- IsolationMonitor (SensorActuatorSWC): [NEW in V2.0]
Continuously measures positive and negative rail isolation impedance to chassis ground.

3. Port Interfaces Specification

3.1 Sender-Receiver Interfaces

- If_CellVoltage:
Carries filtered cell voltage data elements [Array of uint16, mV].
Related Component: CellVoltageMonitor.
- If_CellTemperature:
Carries module temperature readings [Array of sint16, 0.1 deg C].
Related Component: CellTemperatureMonitor.
- If_BatteryStateOfCharge:
Carries calculated SOC percentage and available charge/discharge power limits.
Related Component: SOC_SOE_Estimator.
- If_ThermalCommand:
Transmits cooling pump RPM and chiller valve targets.
Related Component: ThermalManagementService.
- If_CellBalancingCommand: [NEW in V2.0]
Carries PWM target duty cycle and target cell IDs for active balancing.
Related Component: ActiveCellBalancingManager.
- If_IsolationStatus: [NEW in V2.0]
Carries measured isolation resistance in kOhms and fault flag.
Related Component: IsolationMonitor.

3.2 Client-Server Interfaces

- If_ContactorCommand:
Operations: RequestPrecharge(), OpenMainRelay(), EmergencyShutdown(), FastChargeConnect().
Related Component: ContactorControlManager.

4. Component Port Allocations

- ActiveCellBalancingManager:
 - Port R_CellVoltage: Required Port (RPort), Interface: If_CellVoltage
 - Port P_BalancingCmd: Provided Port (PPort), Interface: If_CellBalancingCommand
- IsolationMonitor:
 - Port P_IsolationState: Provided Port (PPort), Interface: If_IsolationStatus
- ContactorControlManager:
 - Port R_BatterySOC: Required Port (RPort), Interface: If_BatteryStateOfCharge
 - Port R_IsolationCheck: Required Port (RPort), Interface: If_IsolationStatus
 - Port P_ContactorControl: Provided Port (PPort), Interface: If_ContactorCommand